Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P06

Past Paper 6

Time: 25 minutes | Paper Reference **PMAT2/N06**

Mathematics

Level 2
Section A (Non-Calculator)



You must have:
Pen, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks
16 /16

My signature confirms that I will not discuss the content of the test with anyone.

Signature: ~~Master Functional Skills (MFS)~~

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Write your final answers in the boxes provided.
- Answer the questions in the spaces provided – *there may be more space than you need.*
- You **must** show clearly how you get your answers in the spaces provided. Marks will be awarded for your working out.
- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may not be used**.
- Take the value of π to be 3.14

Information

- The total mark for this section is 16.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

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SECTION A

Answer ALL questions. Write your answers in the spaces provided.

1 Jai needs to buy 25 lollipops for a party.
He sees this offer.



Jai knows a normal selection bag contains 18 lollipops.
He thinks he will have enough lollipops if he buys a selection bag with this offer.

Is Jai correct?
Show why you think this.

(3)

Step 1: Calculate the extra lollipops (1/3 of 18)

Extra lollipops = 18 ÷ 3 = 6 lollipops

$$3\overline{)18} = 6$$

[M1: 18 ÷ 3 = 6 extra lollipops]

Step 2: Calculate total lollipops in the offer bag

Total = 18 (normal) + 6 (extra) = 24 lollipops [M1: 18 + 6 = 24]

Step 3: Compare with the 25 lollipops required

Jai needs 25 lollipops for the party.

24 < 25 (He is 1 lollipop short: 25 − 24 = 1)

Conclusion: Jai is NOT correct (he only has 24 lollipops) [A1: No & 24]

• Tutor Method Note: Alternative multiplier is 18 × 4/3 = 72/3 = 24. Since 24 < 25, he falls short. ✓

No (only gets 24 lollipops)

(Total for Question 1 is 3 marks)

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2 Here is a formula

$$d = \frac{180\,(n-2)}{n}$$

Find the value of d when $n = 5$

(3)

Step 1: Substitute n = 5 into the formula

$$d = \frac{180\,(5-2)}{5}$$

Step 2: Calculate inside brackets (BIDMAS)

$5 - 2 = 3$ -> $d = (180 \times 3) \div 5$ [M1: 180 × 3]

Step 3: Multiply 180 by 3

2 (carry 2 tens from 3 × 8 = 24)

$$\begin{array}{r} 180 \\ \times\ \ 3 \\ \hline 540 \end{array}$$

180 × 3 = 540

Step 4: Short division (540 ÷ 5)

$$5\overline{)540} = 108$$

• 5 ÷ 5 = 1, 4 ÷ 5 = 0 r 4, 40 ÷ 5 = 8 [M1 A1: d = 108]

• **Geometric Context & Quick Check:**
This formula gives the interior angle of a regular polygon with n sides (n=5 is a regular pentagon: 108°).

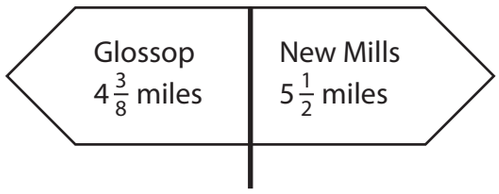
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(Total for Question 2 is 3 marks)

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3 Andrew is a member of a walking club.

He sees this sign next to a footpath.



Andrew will walk from this sign along the footpath to Glossop to meet his friend. They will then both walk back along the footpath to the sign and then onto New Mills.

Work out the total distance that Andrew walks.

Give your answer as a mixed number.
You **must** show your working.

(4)

Step 1: Break down the walking journey

- Sign to Glossop: 4 3/8 miles
- Glossop back to Sign: 4 3/8 miles
- Sign to New Mills: 5 1/2 miles

Step 2: Add the Glossop out-and-back distance

4 3/8 + 4 3/8 = (4 + 4) + (3/8 + 3/8) = 8 6/8 miles [M1: 8 6/8]

Step 3: Find common denominator to add New Mills (5 1/2)

Convert 1/2 to eighths: 1/2 = 4/8 (× 4 top & bottom) [M1: common denom]

Total Distance = 8 6/8 + 5 4/8

Step 4: Combine whole numbers and fractions

- Whole numbers: 8 + 5 = 13
- Fractions: 6/8 + 4/8 = 10/8 = 1 2/8 = 1 1/4

Total = 13 + 1 1/4 = 14 1/4 miles [M1 A1: 14 1/4]

- **Mixed Number Rule:**

Improper fraction method: 35/8 + 35/8 + 44/8 = 114/8

114 ÷ 8 = 14 remainder 2 -> 14 2/8 = 14 1/4 miles.

$14\frac{1}{4}$ miles

(Total for Question 3 is 4 marks)

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4 A box contains bags of crisps.
Each bag of crisps is either beef flavour, prawn flavour or cheese flavour.

Beth is going to take at random a bag of crisps from the box.

The table shows each of the probabilities that the flavour will be beef or will be cheese.

flavour	beef	prawn	cheese
probability	0.4		0.35

0.25

(a) Work out the probability that Beth takes a bag of prawn flavour crisps.

(2)

Key Rule: Sum of all mutually exclusive probabilities = 1

P(beef) + P(cheese) + P(prawn) = 1.00

Step 1: Add the known probabilities (beef + cheese)

$$\begin{array}{r} 0.40 \\ +\ 0.35 \\ \hline 0.75 \end{array}$$

P(beef or cheese) = 0.75

[M1: 0.4 + 0.35 = 0.75]

Step 2: Subtract from 1 to find P(prawn)

borrows: 0 9 10

$$\begin{array}{r} 1.00 \\ -\ 0.75 \\ \hline 0.25 \end{array}$$

P(prawn) = 1 − 0.75 = 0.25

[A1: 0.25 oe]

• Check: 0.40 + 0.35 + 0.25 = 1.00 (verified) ✓ Equivalent: 1/4 or 25%

0.25

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200 workers are asked about the favourite drink they have at work.

Some of the results are shown in the table below.

(b) Complete the two-way table. **(2)**

Part (b) Two-Way Table Calculations:

1) Office Water: Total water − Warehouse = 27 − 10 = 17

2) Warehouse Coffee: Total coffee − Office = 46 − 8 = 38

3) Total Tea: Grand total − Water − Coffee = 200 − 73 = 127

4) Office Tea: Office total − Water − Coffee = 88 − 25 = 63

Check totals: 63 + 64 = 127 (verified) / 10 + 64 + 38 = 112 (verified) [M1 A1: 4 table values]

		Favourite drink			
		water	tea	coffee	total
Workers	office	17	63	8	88
	warehouse	10	64	38	112
	total	27	127	46	200

(c) What is the probability that a worker choosing coffee works in the office?
Give your answer as a fraction in its simplest form. **(2)**

Part (c) Probability that a coffee drinker works in the office:

- Condition: Worker chooses coffee -> Total coffee drinkers = 46 (denominator)
- Number of office workers who drink coffee = 8 (numerator)

Probability $= \frac{8}{46} \overset{\div 2}{=} \frac{4}{23}$ (simplest form: 23 is a prime) [M1 A1: 4/23]

$\frac{4}{23}$

(Total for Question 4 is 6 marks)

TOTAL FOR SECTION A IS 16 MARKS

Please check the examination details below before entering your candidate information

Candidate surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-P06

Past Paper 6

Time: 1 hour 30 minutes | Paper Reference **PMAT2/C06**

Mathematics

Level 2
Section B (Calculator)



You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, pair of compasses. Tracing paper may be used.

Total Marks: 48 /48

My signature confirms that I will not discuss the content of the test with anyone.

Signature: ~~Master Functional Skills (MFS)~~

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
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- Check your working and answers at each stage.
- Diagrams are **not** accurately drawn, unless otherwise indicated.
- **Calculators may be used.**
- If your calculator does not have a π button take the value of π to be 3.14

Information

- The total mark for this section is 48.
- The total mark for this paper is 64.
- The marks for **each** question are shown in brackets.
  – *use this as a guide as to how much time to spend on each question.*
- This sign ☑ shows where marks will be awarded for showing your checks.

Advice

- Read each question carefully before you start to answer it.
- Check your answers if you have time at the end.

Turn over ▶

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SECTION B

Answer ALL questions. Write your answers in the spaces provided.

1 Olivia plans to drive from Birmingham to Manchester non-stop.

The distance is 87 miles.
She has allowed 1.5 hours for the journey.

(a) What is the average speed she must drive to complete the journey in 1.5 hours?

(2)

Part (a) Average Speed Calculation:

Formula: Average Speed = Distance ÷ Time

Speed = 87 miles ÷ 1.5 hours = 58 mph [M1: 87 ÷ 1.5]

```
   5 8
15 ) 8 7 0
```

• 87 ÷ 1.5 = 870 ÷ 15 = 58 mph [A1: 58]

58 mph

☑ (b) Use a reverse calculation to show a check of your answer.

(1)

Part (b) Reverse Calculation Check:

Reverse check formula: Speed × Time = Distance

58 mph × 1.5 hours = 87 miles (verified) [B1: check]

(Total for Question 1 is 3 marks)

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2 Brian is a sports reporter.

Team A played 30 rugby matches.

The table shows information about the number of tries team A scored in these matches.

number of tries	number of matches	
0	3	
1	7	
2	11	
3	9	
total	30	

0
7
22
27
56

(a) Work out the mean number of tries team A scored per match.

(2)

Part (a) Work out total tries and mean per match:

Total Tries = (0×3) + (1×7) + (2×11) + (3×9) = 56 tries

Mean = 56 ÷ 30 = 1.866... ≈ 1.87 tries per match [M1 A1: 1.87]

1.87 tries

Brian wants to compare two teams.
He knows

- team A had a range of 3 tries
- team B had a range of 4 tries.

(b) Which team is more consistent scoring tries?
Explain why you think this.

(1)

Part (b) Team A is more consistent because its range is lower (3 < 4).

A smaller range indicates less variation / greater consistency in scores. [B1: Team A lower range]

(Total for Question 2 is 3 marks)

3 The head teacher at a school is organising for some year 7 and some year 8 pupils to go on a school trip.
72 people in total will go on the school trip.

There will be 1 adult to every 5 pupils.

The ratio of the number of year 7 pupils to the number of year 8 pupils will be 3 : 1

How many adults, year 7 pupils and year 8 pupils will go on the trip?

(4)

Step 1: Divide 72 people into adults & pupils (Ratio 1 : 5)

- Total ratio parts = 1 (adult) + 5 (pupils) = 6 parts
- **1 part = 72 ÷ 6 = 12 people** [M1: 72 ÷ 6 = 12]

Number of Adults = 1 × 12 = 12 adults

Number of Pupils = 5 × 12 = 60 pupils (or 72 − 12 = 60)

Step 2: Divide 60 pupils into Year 7 & Year 8 (Ratio 3 : 1)

- Total pupil ratio parts = 3 (Yr 7) + 1 (Yr 8) = 4 parts
- **1 pupil part = 60 ÷ 4 = 15 pupils** [M1: 60 ÷ 4 = 15]

Year 8 pupils = 1 × 15 = 15 pupils

Year 7 pupils = 3 × 15 = 45 pupils [M1: 15 × 3 = 45]

- **Full Cross-Check:** [A1: All 3 correct]

12 adults + 45 Year 7 + 15 Year 8 = 72 people total (verified) · Ratio 45:15 = 3:1 (verified)

12	adults
45	year 7 pupils
15	year 8 pupils

(Total for Question 3 is 4 marks)

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4 Josh drives a lorry.
He is planning his route.

On the planned route there is a low bridge.
The maximum height of a lorry allowed under the bridge is 14 feet.

Josh knows the lorry is 4.2 m high.

Josh uses 1 foot = 0.3048 m.

(a) Will the lorry be allowed under the bridge?
Show why you think this.

(2)

Part (a) Convert bridge height to metres and compare:

Bridge max height = 14 feet × 0.3048 m/foot = 4.2672 m [M1: conversion]

4.2000 m (lorry) < 4.2672 m (bridge limit)

Lorry is 0.0672 m (6.72 cm) lower than bridge. Yes, it will pass. [A1: Yes]

Yes (4.2 m < 4.2672 m)

☑ (b) Use a reverse calculation to show a check of your answer.

(1)

Part (b) Reverse calculation check:

Reverse calculation check: 4.2672 m ÷ 0.3048 = 14 feet (verified)

[B1: check]

(Total for Question 4 is 3 marks)

5 Yasmine invests £4000

For the first 2 years she receives annual compound interest of 3%
In year three she receives annual compound interest of 2.5%

At the end of year three Yasmine wants to buy a car for £4500

She will use all of the investment and interest towards the cost of the car.

Work out how much more money Yasmine needs to buy the car.
You **must** show your working

(5)

Step 1: First 2 years at 3% annual compound interest

- Interest multiplier = 1 + 0.03 = 1.03
- **End of Year 1: £4,000 × 1.03 = £4,120.00** [M1: 4000 × 1.03]
- **End of Year 2: £4,120 × 1.03 = £4,243.60** [M1: 4000 × 1.03^2]

Step 2: Year three at 2.5% annual compound interest

- Year 3 multiplier = 1 + 0.025 = 1.025
- **End of Year 3: £4,243.60 × 1.025 = £4,349.69** [M1: 4243.6 × 1.025]

Step 3: Calculate remaining money needed for £4,500 car

Money needed = £4,500.00 − £4,349.69

```
  4500.00
- 4349.69
----------
   150.31
```

Extra Money Needed = £150.31

[M1 A1: £150.31]

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Question 5 Summary & Financial Review Card:

- **Breakdown of Yasmine's Investment Growth:**

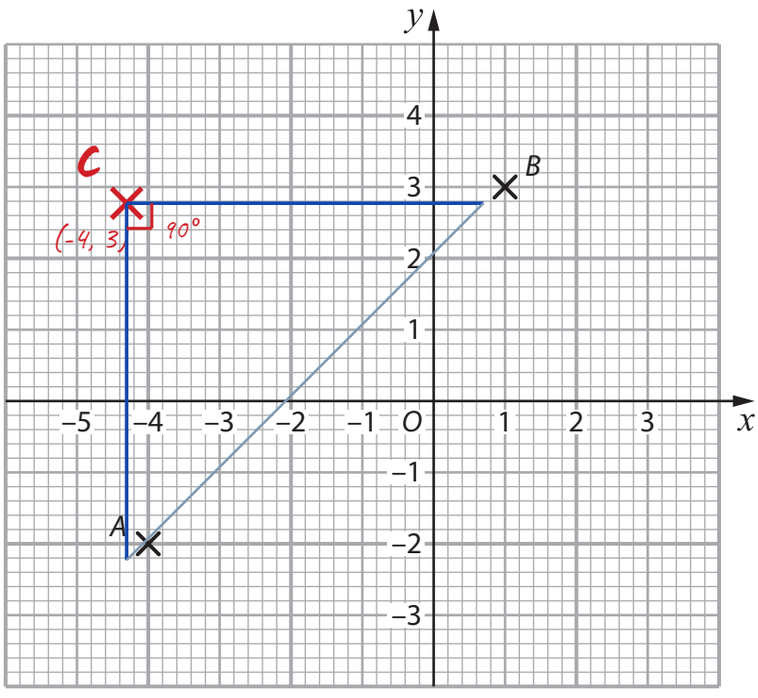
1. Starting Principal: £4,000.00
2. After 2 Years at 3% compound: £4,243.60 (+£243.60 earned)
3. After Year 3 at 2.5% compound: £4,349.69 (+£106.09 earned)
4. **Total Interest Earned: £349.69 (towards £4,500 car target)**

Yasmine is only £150.31 away from buying her car in full!

£ 150.31

(Total for Question 5 is 5 marks)

6



(a) Write down the coordinates of point *A*.

(1)

Part (a) Coordinates of Point A: Read x = −4, y = −2

(**−4**, **−2**) [B1]

(b) Mark with a cross the point *C* on the grid so that angle *ACB* is a right angle.

(1)

Part (b) Plot Point C so angle ACB = 90° (Right Angle):

- Point C at (−4, 3) gives vertical line AC and horizontal line BC. [B1: C at (-4, 3)]
- Perpendicular lines meet at 90°, forming right angle at C.

(c) Write down the sum of the interior angles of a triangle.

(1)

Part (c) Sum of interior angles of ANY triangle = 180°

180 [B1] °

(Total for Question 6 is 3 marks)

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7 Carlos records the number of tweets his company posts every year on social media.

year	2014	2015	2016	2017	2018	2019
number of tweets	452	325	744	1022	712	750

(a) Find the median number of tweets.

(2)

Part (a) Find the median number of tweets:

Step 1: Order the 6 values from smallest to largest:

325, 452, 712, 744, 750, 1022

Median = (712 + 744) ÷ 2 = 1456 ÷ 2 = 728 *[M1 A1: 728]*

728

In 2019, 6% of the tweets were about job vacancies.

(b) How many of the tweets in 2019 were about job vacancies?

(2)

Part (b) Work out 6% of the 750 tweets in 2019:

· *In 2019, total tweets = 750 · 6% = 6 ÷ 100 = 0.06*

Number of tweets = 750 × 0.06 = 45 tweets *[M1 A1: 45]*

Check: 10% = 75, 1% = 7.5 -> 6% = 7.5 × 6 = 45 (verified)

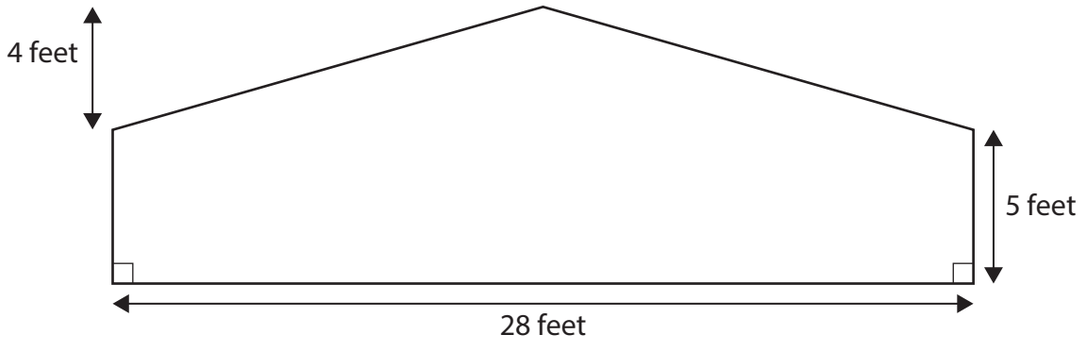
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(Total for Question 7 is 4 marks)

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8 Myra works as a volunteer at her local pony club.
She is going to cover the front of the stables with paint.

The diagram shows a sketch of the front of the stables.



Myra will buy the paint she needs.

1 tin of paint

- covers 7 m^2
- costs £6.45

She uses this rule.

Area in square feet ÷ 10.764 = area in square metres.

Work out the total cost of the tins of paint Myra will buy

(6)

Step 1: Calculate the total front area in square feet

- Rectangle area = 28 × 5 = 140 sq ft [M1: 28 × 5 = 140]
- Gable area = (28 × 4) ÷ 2 = 112 ÷ 2 = 56 sq ft [M1: (28×4)/2 = 56]

Total Area = 140 + 56 = 196 sq ft [M1: 196 sq ft]

Step 2: Convert square feet to square metres

Rule: Area in sq ft ÷ 10.764 = Area in sq metres

Area = 196 ÷ 10.764 = 18.2088... m^2 [M1: 196 ÷ 10.764]

-> Continued on next page for paint tins calculation & total cost...

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Step 3: Calculate the number of paint tins needed

- 1 tin covers 7 m^2
- **Number of tins = 18.2088 ÷ 7 = 2.60 tins** [M1: 18.208 ÷ 7]

Rounding Rule: Paint is sold in whole tins -> Round UP to 3 tins

Step 4: Calculate total cost of the paint

Total Cost = 3 tins × £6.45 = £19.35 [M1 A1: £19.35]

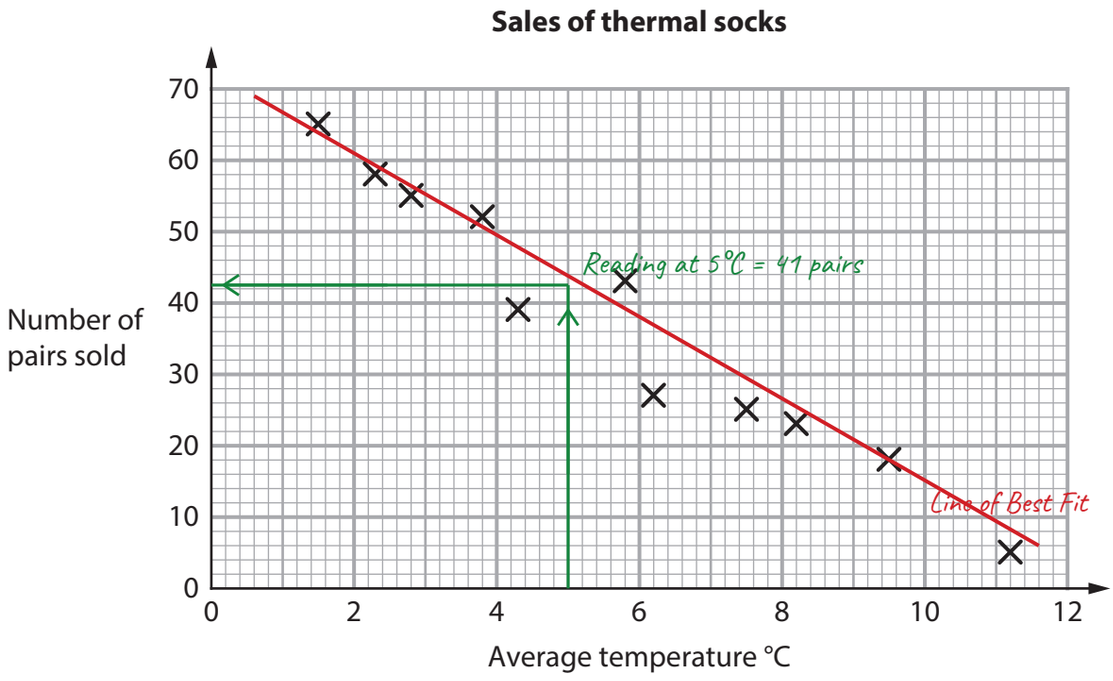
- **Key Functional Insight on Rounding Up:**
  1. 2 tins would only cover 2 × 7 = 14 m^2, leaving 4.21 m^2 unpainted!
  2. Therefore, Myra MUST purchase 3 full tins of paint.
  3. **Final cost = 3 × £6.45 = £19.35 (verified)** ✓

£ 19.35

(Total for Question 8 is 6 marks)

9 Leona works in a sock shop.

The scatter diagram shows the number of pairs of thermal socks sold and the average temperature that month.



(a) Draw a line of best fit onto the scatter diagram.

(1)

[Line of best fit drawn on scatter diagram above]

[B1: Line drawn]

(b) Describe the correlation between the number of pairs of thermal socks sold and the temperature.

(1)

Negative correlation *(as temperature increases, thermal sock sales decrease)*

[B1: Negative correlation]

Next month the predicted average temperature is 5°C
Leona needs to estimate how many pairs of thermal socks she will sell.

(c) Estimate the number of pairs of thermal socks she will sell next month.

(1)

From scatter graph at 5°C, following projection line:

Estimated sales = 41 pairs

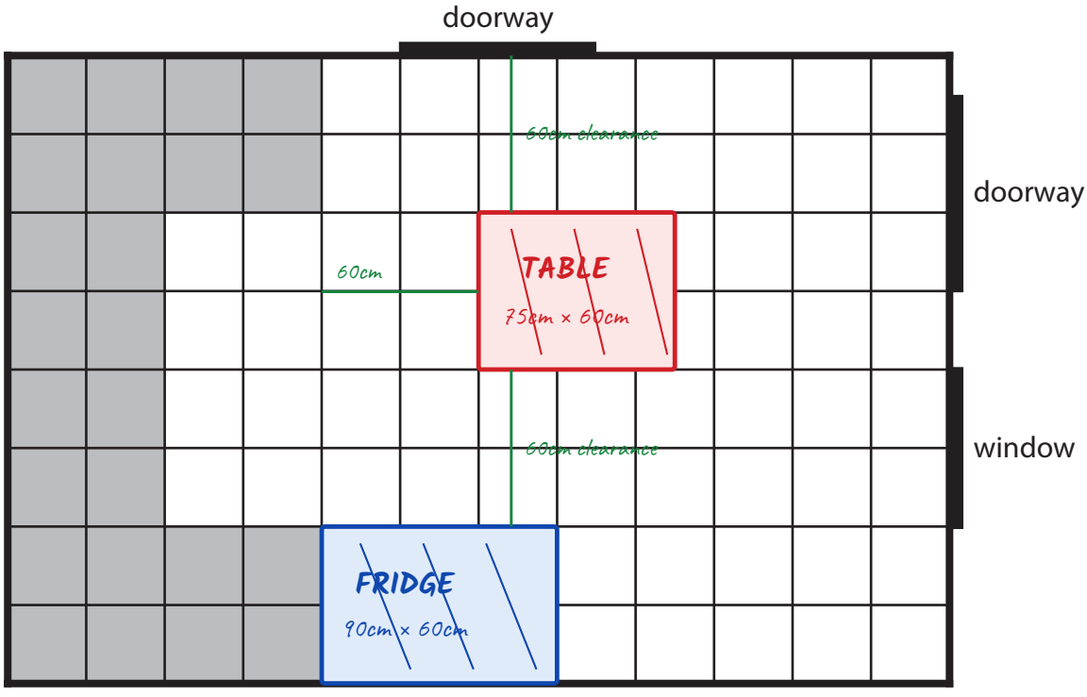
[B1: 41 pairs]

41

(Total for Question 9 is 3 marks)

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10 Chris has moved to a new flat.
The grid shows a plan of the kitchen.



Key

walls

cupboards

1 square length on the grid represents 0.3 m in the kitchen

Chris needs to put a fridge and a table in the kitchen.

The fridge will need a rectangular space 90 cm by 60 cm with the longest side against a wall but not against a window.

The table will need a rectangular space 60 cm by 75 cm and to be at least 60 cm away from the doorways, the cupboards, the fridge and not against a wall.

Draw the space for the fridge and the space for the table on the grid for Chris.
Remember to label the items.

(3)

• Fridge: 3 × 2 sq (against wall) • Table: 2.5 × 2 sq • ≥ 2 sq clearance [M1 M1 A1]

(Total for Question 10 is 3 marks)

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11 Nigel is the secretary of a football club.

He pays three match officials each week.

He has this information for the payments he made for the last 5 weeks.

Week	1	2	3	4	5
Payment made (£)	62.94	47.12	92.37	74.80	81.45

The table shows the match fees and expenses the officials will receive in week 6

official	**fee**	**expenses**
referee	£36	46 miles at 30p per mile
assistant 1	£27	14 miles at 30p per mile
assistant 2	£27	23 miles at 30p per mile

The total payment for each official is made up of a match fee and expenses.
Nigel pays 67% of the total payments for these three match officials.

Nigel thinks the payment he makes in week 6 is more than the median payment he made for the previous 5 weeks.

Is Nigel correct?
Show why you think this.

(5)

Step 1: Find the median payment of the previous 5 weeks

Order 5 past payments: £47.12, £62.94, £74.80, £81.45, £92.37

Median payment = £74.80 (the middle value) [M1: £74.80]

Step 2: Calculate total fees and expenses for Week 6

- Match Fees: £36 (ref) + £27 (ast 1) + £27 (ast 2) = £90.00
- Total Mileage: 46 miles + 14 miles + 23 miles = 83 miles
- Expenses: 83 miles × £0.30 per mile = £24.90 [M1: £24.90 expenses]

Total Officials Payment = £90 + £24.90 = £114.90 [M1: £114.90 total]

-> Continued on next page for Nigel's share and comparison...

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Step 3: Calculate Nigel's share (67% of total payment)

Nigel pays: £114.90 × 0.67 = £76.98 [M1: 0.67 × 114.90]

Step 4: Compare Week 6 payment with the median:

- Comparison of Week 6 vs Previous 5 Weeks:
  1. Nigel's Week 6 Payment: £76.98
  2. Median of Previous 5 Weeks: £74.80
  3. Comparison: £76.98 > £74.80 (by £2.18 more)

Conclusion: Yes, Nigel is correct. His Week 6 payment (£76.98) is more than the median payment (£74.80). [A1: Yes & figures]

Yes (£76.98 > £74.80)

(Total for Question 11 is 5 marks)

12 Tammy wants to make chocolate sweets.

The sweets will be solid chocolate in the shape of a sphere.
Each sweet will have a radius of 2 cm.

Tammy will melt chocolate blocks to make the sweets.
Each chocolate block is a cuboid 19 cm by 14 cm by 0.75 cm.

She has this formula

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3$$

where r = radius
$\pi = 3.14$

Tammy wants to make 45 sweets.
She thinks 7 blocks of chocolate are enough to make 45 sweets.

Is Tammy correct?
Show why you think this.

(6)

Step 1: Calculate volume of 1 spherical sweet

Formula: Volume = 4/3 π r^3 (r = 2 cm, π = 3.14)

$r^3 = 2^3 = 8$ cm³ [M1: $2^3 = 8$]

Volume = 4/3 × 3.14 × 8 = 33.493 cm³ [M1: 33.493 cm^3]

Step 2: Total volume of chocolate needed for 45 sweets

Total Volume = 45 × 33.49 cm³ = 1,507.20 cm³ [M1: 1507.2 cm^3]

Step 3: Calculate the volume of 1 cuboid chocolate block

Volume = length × width × height = 19 × 14 × 0.75

Volume of 1 block = 199.50 cm³ [M1: 199.5 cm^3]

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Step 4: Compare 7 blocks with the volume required:

- **Method A (Volume Comparison):**
  - Volume in 7 blocks: $7 \times 199.50\ cm^3 = 1{,}396.50\ cm^3$
  - Volume needed: 45 sweets = $1{,}507.20\ cm^3$
  - **$1{,}396.50\ cm^3 < 1{,}507.20\ cm^3$ (Deficit of $110.70\ cm^3$)**
- **Method B (Blocks Required):** [M1: compare]
  - **$1{,}507.20 \div 199.50 = 7.554...$ blocks (needs 8 full blocks)**

Conclusion: Tammy is NOT correct.

7 blocks produce only 1,396.5 cm^3, which is not enough for 45 sweets.

She needs 7.55 blocks (8 whole blocks of chocolate). [A1: No & figures]

No (needs 7.55 blocks / 8 blocks)

(Total for Question 12 is 6 marks)

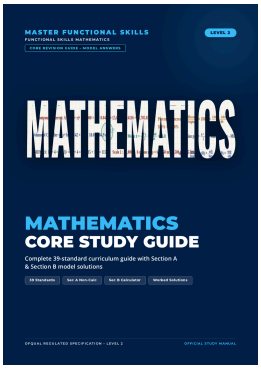
TOTAL FOR SECTION B IS 48 MARKS
TOTAL FOR PAPER IS 64 MARKS

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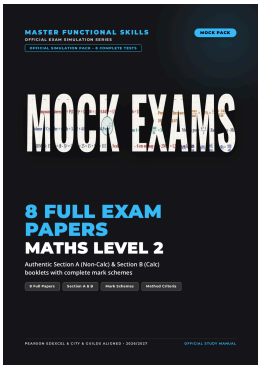


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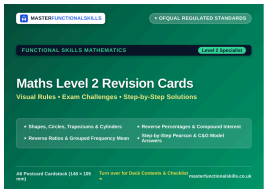


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